

COMPUTER HARDWARE FUNDAMENTAL

MODULE 01: INTRODUCTION

The PC Hardware

- PC Development Timeline
- System Components
- PC Case
- System Types
- Power Supply
- Switches and Connectors

PC Development Timeline

Year Released	PC
1972	8088
1973	Micral
1974	8H
1976	Apple I
1977	Apple II
1979	6800
1981	Star IBM's 1 st PC
1983	Apple Lisa
1984	Macintosh
1986	DeskPro 386

PC Development Timeline

Year Released	PC
1987	PS/2
1988	EISA
1989	486
1993	P5
1995	P6
1997	Pentium II K6
1999	Pentium III
2000	1GHz processors (AMD Duron)
2001	Pentium 4
2006	P6 & Netburst / Core Architecture
2008	Penryn Microarchitecture

System Components

- Microprocessor
 - Central Processing Unit (CPU)
 - Central to all processing done by the computer
 - Reads and writes data and instructions to and from storage devices and performs calculations and other data processing

System Components

- Port
 - Physical connector that allows a cable from a peripheral device to be attached to the computer.

System Components

- Motherboard
 - Core of the system which controls all the devices



System Components

- Processor
 - Serves as the brain or engine of the PC



System Components

- Memory
 - Contains data which the processor is using at a given time



System Components

- Hard Disk Drives
 - Secondary storage that is used when large amounts of data have to be stored



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System Components Monitor

- Video display which shows information



System Components

- Video Card
 - Controls the information displayed on the monitor



System Components

- Power Supply
 - Source of electrical power of every part of the PC



PC Power Supply

- Provides all of the different voltages the computer needs to operate properly.
- Converts the power received from a standard power outlet, converts AC to DC.
- Designed to give an efficient and inexpensive power source, as well as it should generate a minimum amount of heat.

System Components

- Case
 - Frame that houses the system unit



PC Case Functions

- Housing
 - The case usually makes room for the power supply unit, expansion slots/bays, and I/O ports connected to the motherboard.

PC Case Functions

- Ventilation
 - The case should be well ventilated and can be equipped with cooling fans.

PC Case Functions

- EMF Elimination and Grounding
 - The case can keep internal components from generating EMF to a minimum.
 - The case can also provide adequate grounding, causing immunity to static electric discharges.

PC Case Form Factors

- Form Factors
 - Refers to the shape and general physical layout of a component.

PC Case Form Factors

Types:

- ATX
- micro ATX
- Extended ATX
- ITX

PC Case Forms Factors

- ATX
 - Advanced Technology Extended. This case can house ATX, m-ATX, and ITX motherboards. Can usually fit 7 or more 120mm fans.



PC Case Forms Factors

- micro-ATX
 - A smaller ATX PC Case that can house m-ATX or ITX motherboards. This case can usually fit only up to 7 120mm fans.



PC Case Forms Factors

- extended-ATX
 - e-ATX cases house mostly all of ATX sized motherboards (ATX, e-ATX, m-ATX, and ITX). These cases provide more HDD/SSD SATA holders than ATX cases and can hold more than 7 case fans. Commonly used on enthusiast-level workstation (water-cooled systems, etc.)



PC Case Forms Factors

- ITX
 - ITX cases are the smallest PC Case form factor. It can only support ITX motherboard and is extremely limited on ventilation.



Obsolete PC Case Form Factors

- AT
 - 1984, with a round hole at the back used for inserting the DIN plug of old keyboards.



Obsolete PC Case Form Factors

- Baby-AT
 - Scaled down model of AT



Obsolete PC Case Form Factors

- LPX
 - Low profile extension format

